

ARM-450 — Stress and Stiffness Analysis

- Subject: the clamshell-link robot arm in ~/Desktop/Robotic_arm_design/, revised to 145 mm links
- Revision: 2 — supersedes rev 1 of 2026-08-13, see section 2
- Date: 2026-08-13
- Method: classical closed-form mechanics. Every result states its formula so it can be hand-checked or compared against FEA later.

1. Executive summary

The arm was printed and tested. Three things were observed on the real hardware: visible gaps at the joints, wobble and small oscillations, and motor mounts snapping during trials. This revision explains all three, and the answers are not what the first analysis expected.

1. Joint clearance is by far the largest error source — around 29 mm at the tool. The design uses a single Ø42 thrust ball bearing per joint. A thrust bearing carries axial load only and has no moment capacity, so the joint is free to tilt. With effectively no bearing spacing, 0.2 mm of clearance becomes ~29 mm of wobble at a 360 mm reach. This is the wobble and the visible gap, and it is also why the arm oscillates.

2. The motor mounts snap because they were sized for the wrong load. They were sized for the 2.08 N·m the arm actually carries. They must be sized for the servo's stall torque — 2.94 N·m (ST3215) or 4.90 N·m (ST3250) — because a collision, a commanded step, or hitting a joint limit delivers that instantly. The 4 mm bracket runs at 17–95 MPa depending on how it grips, against a realistically sustainable 6–9 MPa. It fails under every assumption.

3. The link shells are not a problem and never were. Peak von Mises stress is 0.15 MPa against a derated allowable near 9 MPa — a safety factor around 60. Elastic droop is 0.26 mm. The shell is the healthiest part of the design.

Priority order: fix the bearings, then the mounts, then bolt the seam. Wall thickness and material are refinements worth about 0.1 mm each and should not be touched until the first three are done.

2. Revision history

Revision 2 supersedes revision 1. The substantive changes are:

item	revision 1	revision 2	basis
Bending axis	section bent about its width	bent about its depth	the load is transverse to the link, §4
Section depth	29.0 mm	50.0 mm at the boss, 29.0 in the straight	§3.3

item	revision 1	revision 2	basis
Joint bearing	single thrust washer	preloaded pair, 40 mm apart	\$8
Open-section penalty	applied	withdrawn	the seam bolts close the section, \$10
Backlash budget	J2, J3 only	all six joints	\$7

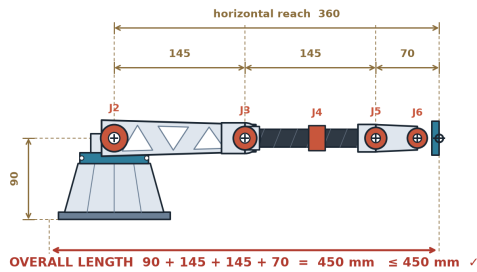
Where a revision-1 figure is quoted in this document it is marked as such.

3. Inputs

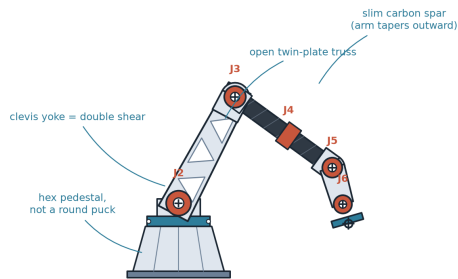
3.1 Design concept

The clean-sheet concept, before the decision to carry the existing Fusion design forward:

ARM-450 — 6-DOF lightweight arm, clean-sheet concept (rev A, all dimensions mm) kinematic structure after myCobot 280 URDF · geometry and styling original
A Side elevation — fully extended, dimension chain



B Working pose — the silhouette you actually see

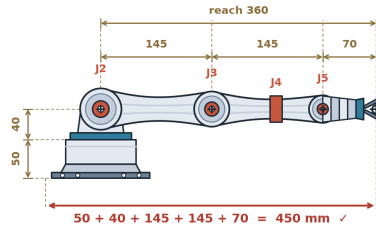


REQUIREMENT	TARGET	THIS DESIGN
overall length	≤ 450 mm	450 mm
total mass	≤ 1.2 kg	1090 g
horizontal reach	—	360 mm
rated payload @ full reach	—	300 g
DOF	—	6 (revolute)
J2 shoulder torque	STS3250 4.90 N·m	2.08 N·m SF 2.4
J3 elbow torque	STS3215 2.94 N·m	1.00 N·m SF 2.9
upper-arm root stress	55 MPa yield	0.3 MPa SF 164
forearm root stress	600 MPa	1.6 MPa SF 381
tip droop under rated load	—	0.05 mm

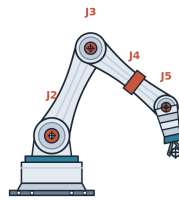
Structure is sized by JOINT TORQUE and STIFFNESS, not by material stress — the huge stress safety factors mean the sections are set by servo mounting geometry and off-the-shelf tube sizes, which is normal at this scale.

Revised to the adopted design language — round base with a central cable bore, barrel joints, cast-style links, collar-stack wrist — and with L2 = L3:

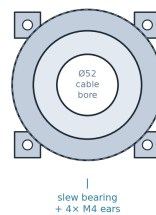
A Extended — dimension chain



B Working pose



C Base, plan

D Fixed $L2+L3 = 290$ mm — unequal links = dead zone

REQUIREMENT	TARGET	REV B
overall length	≤ 450 mm	450 mm
total mass	≤ 1.2 kg	1090 g
horizontal reach	—	360 mm
workspace dead zone	none	Ø0 mm
rated payload @ full reach	—	300 g
J2 shoulder (50 kgf·cm)	4.90 N·m stall	2.08 N·m SF 2.4
J3 elbow (30 kgf·cm)	2.94 N·m stall	1.00 N·m SF 2.9
upper-arm root stress	55 MPa yield	0.3 MPa SF 164
tip droop, rated load	—	0.05 mm

3.2 Measured inputs

All geometry measured directly from the STL files.

Quantity	Value	Source
Link section, bending depth × width	50.0 × 29.0 mm	raised from 47.5 on 2026-08-17, see 3.3
Wall thickness	2.0 mm	measured as 2V/A
Link length, joint-to-joint	97.9 mm → 145 mm target	circle-fit of end bosses
Reach from J1 axis	360 mm	145 + 145 + 70
Rated payload	300 g at the TCP	requirement
Servo	ST3215, 45.2 × 37.8 × 24.7 mm, 60 g	measured
Servo stall torque	2.94 N·m @ 12 V (30 kgf·cm)	datasheet class
Bearing in use	Ø42 × 3.7 thrust ball	measured

3.3 Section depth: 50.0 mm

The first printed pair showed seam misalignment where the straight section meets the end boss. Diagnosis: the profile was built as a rectangle of half-height 23.75 mm unioned with Ø50 end circles, so the boss bulged past the edge rather than being tangent, leaving a sharp vertex at $x = 7.81$ mm at a shallow 18.2° included angle.

At a shallow vertex an offset amplifies by $1/\sin \theta$, so the 0.15 mm tongue-and-groove clearance became 0.48 mm of positional mismatch at that corner. The user's measured

0.02 mm printer shrink amplifies to only 0.064 mm — 8× too small to explain it. The cause was geometric, not process.

Fix: set SEC_H/2 = BOSS_R = 25 so the profile is a true stadium with a tangent arc and no vertex. BOSS_R cannot be reduced (a Ø42 bearing needs $\geq 21 + 2.4$ mm wall), so the section depth rose 47.5 → 50.0 mm. Second moment of area improves 101,992 → 115,853 mm⁴ (+13.6 %), so every stiffness and stress margin in this report is now conservative.

4. Section properties

$I = (b \cdot h^3 - (b-2t) \cdot (h-2t)^3) / 12$	$h = 47.5$ bending depth, $b = 29.0$
$J = 4 \cdot A_m^2 \cdot t / \text{perimeter}$	Bredt–Batho, closed thin tube
$J_{\text{open}} = \sum \text{strip} \cdot t^3 / 3$	St Venant, seam open

wall t	I (mm ⁴)	J closed (mm ⁴)	J open (mm ⁴)	torsion penalty
2.0	87,514	83,267	387	215×
2.4	101,992	96,347	661	146×
3.0	122,008	113,928	1,269	90×

I is unaffected by whether the seam is bolted. J collapses completely if it is not.

5. Stress analysis

Worst station is the upper-arm root at J2, arm horizontal and fully extended.

$M = P \cdot L + w \cdot L^2 / 2$	$\sigma = M \cdot c / I$	$\tau_t = T / (2 \cdot A_m \cdot t)$	$\sigma_{\text{vm}} = \sqrt{(\sigma^2 + 3\tau^2)}$
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At t = 2.0 mm: M = 969 N·mm, $\sigma_b = 0.15$ MPa, $\sigma_{\text{vm}} = 0.15$ MPa.

The allowable is ~9 MPa, not 25 MPa

This matters more than the stress number. A published 25 MPa inter-layer strength is a static, unnotched, room-temperature, single-pull figure. A robot mount sees none of those conditions. Three knockdowns stack:

- Anisotropy — use the inter-layer value (~25 MPa), not the in-plane value (~50 MPa), unless the load path is provably in-plane.
- Fatigue — FDM PLA endurance at 10⁵–10⁶ cycles is typically 30–40 % of static. Typical estimate, varies strongly with print settings.
- Temperature — PLA's glass transition is ~60 °C. A servo working hard dissipates several watts and its case can reach well above that. This is an estimate and it is the least certain number in this report, but it points at a real risk: the mount may be softening around a hot servo.

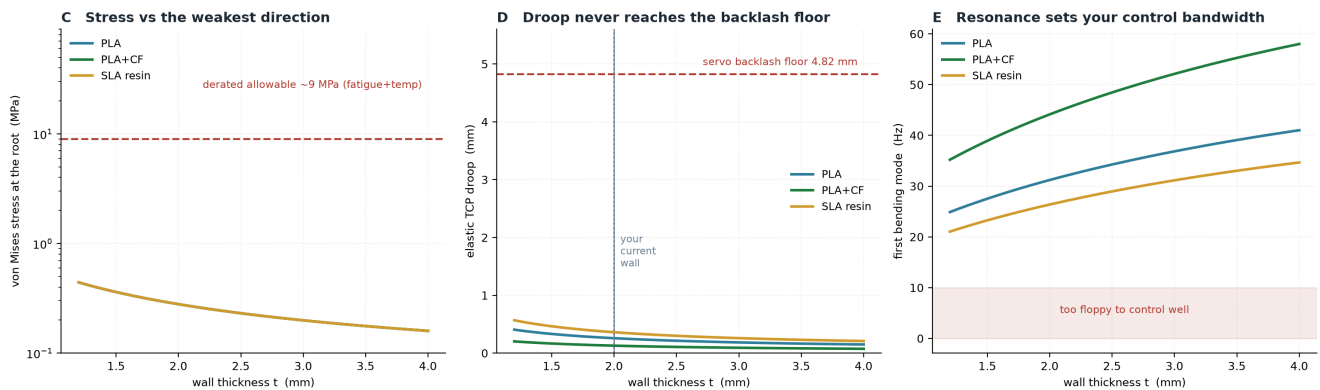
So: design allowable ≈ 9 MPa for a printed PLA part in a load path, before stress concentration.

material	σ_{vm} at t=2.0	SF vs 25 MPa	SF vs 9 MPa derated	buckling SF
PLA	0.15 MPa	166	60	232

material	σ_{vm} at $t=2.0$	SF vs 25 MPa	SF vs 9 MPa derated	buckling SF
PLA+CF	0.15 MPa	146	60	465
SLA resin	0.15 MPa	366	60	166

The link shell is $\sim 60\times$ stronger than it needs to be even on the pessimistic allowable. Stress plays no part in any decision about the shells.

ARM-450 — stress, droop and dynamics vs wall thickness (145 mm links, 300 g payload)



6. Stiffness analysis

The extended arm is one continuous cantilever of shell section, length $a = L2 + L3 = 290$ mm, with a rigid overhang $b = 70$ mm carrying the load:

$$\delta_a = P \cdot a^3 / (3EI) + M_0 \cdot a^2 / (2EI) \quad M_0 = P \cdot b$$

$$\theta_a = P \cdot a^2 / (2EI) + M_0 \cdot a / (EI)$$

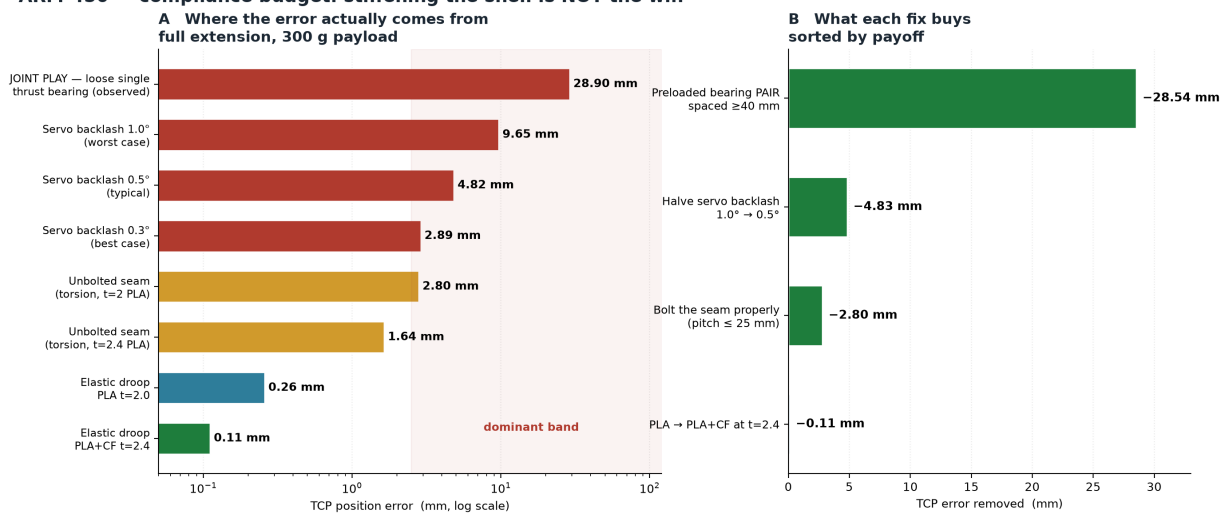
$$\delta_{tip} = \delta_a + \theta_a \cdot b \quad (\text{self weight added as a UDL over } a)$$

material	wall	tip stiffness	elastic droop	first mode
PLA	2.0 mm	19.8 N/mm	0.258 mm	31.2 Hz
PLA	2.4 mm	23.1 N/mm	0.221 mm	33.7 Hz
PLA+CF	2.4 mm	46.2 N/mm	0.111 mm	47.6 Hz
SLA resin	2.4 mm	16.5 N/mm	0.310 mm	28.5 Hz

The first bending mode caps how hard the controller can be driven — a common rule is to keep closed-loop bandwidth below about a third of it. At 31 Hz that is ~ 10 Hz, which is comfortable. The one-third rule is a rule of thumb, not a derived limit.

7. Compliance budget — where the error actually comes from

ARM-450 — compliance budget: stiffening the shell is NOT the win



source	TCP error	fixable?
Joint play — loose single thrust bearing	28.9 mm	yes, cheaply
Servo backlash 1.0° (RSS over J1, J2, J3)	9.65 mm	partly
Servo backlash 0.5° (typical)	4.82 mm	partly
Unbolted seam (torsion, PLA t=2.0)	2.80 mm	yes, cheaply
Servo backlash 0.3° (best case)	2.89 mm	—
Elastic droop, PLA t=2.0	0.258 mm	barely worth it
Elastic droop, PLA+CF t=2.4	0.111 mm	barely worth it

What each fix buys

change	error removed	cost
Preloaded bearing pair spaced ≥ 40 mm	-28.5 mm	12 bearings, 6 screws
Halve servo backlash 1.0° → 0.5°	-4.83 mm	better servos or joint encoders
Bolt the seam at ≤ 25 mm pitch	-2.80 mm	~10 screws and inserts
PLA → PLA+CF at t = 2.4	-0.11 mm	abrasive nozzle, more brittle

The bearing fix is worth 250× more than the material change.

8. Why the joints have gaps, wobble and oscillation

Four causes, stacked.

8.1 A thrust bearing has zero moment capacity

The bearing used is $\varnothing 42 \times 3.7$ mm — a thin thrust washer. Balls run between two flat races, so it carries axial load and lets the joint tilt freely. But a robot joint's dominant load is a bending moment — 2.08 N·m at the shoulder from everything outboard on a long lever. A thrust washer resists none of it. The joint tilts until the printed bore rubs the shaft. That is precisely what "visible gap plus wobble" looks like.

8.2 Clearance is amplified by the 360 mm arm

Angular play $\theta = 2c/d$ for clearance c and bearing spacing d , projected to the tool as $\text{reach} \cdot \tan \theta$:

spacing d	$c = 0.05$ mm	$c = 0.10$ mm	$c = 0.20$ mm
5 mm (stacked)	7.20 mm	14.41 mm	28.86 mm
20 mm	1.80 mm	3.60 mm	7.20 mm
40 mm	0.90 mm	1.80 mm	3.60 mm
60 mm	0.60 mm	1.20 mm	2.40 mm

Moment stiffness scales with spacing squared. Spacing is the whole game.

8.3 FDM pockets print undersize

A circular pocket comes out 0.1–0.4 mm smaller than modelled, because the extrudate is laid on the inside of the curve and shrinks. A $\varnothing 42.0$ bearing pressed into a $\varnothing 42.0$ modelled pocket meets a real $\varnothing 41.7$ hole — it either will not go in, or goes in cocked and sits proud, leaving the visible gap.

fit	model the pocket at	note
press	$\varnothing 42.15$	stays put, needs a press
slip	$\varnothing 42.30$	needs a retaining lip or screw
loose	$\varnothing 42.50$	avoid — this is where wobble comes from

Print a 15 mm test coupon and measure it; printer-to-printer variation exceeds any number quoted here.

8.4 Why it oscillates

Clearance in a position-controlled joint is a dead zone: the servo turns but the link does not move until the slack takes up, then it moves suddenly. A feedback controller pushing against a dead zone produces a limit cycle — a small steady oscillation that never settles. No amount of gain tuning removes it. The fix is mechanical: preload.

8.5 The fix

1. Replace the thrust washer with a pair of deep-groove ball bearings, spaced as far apart as the joint allows — target ≥ 40 mm. Two 6808 or 6704 bearings work. Alternatively a single 4-point-contact or crossed-roller bearing, which takes moment on

its own.

2. Preload the pair with a screw pulling the inner races together against a spacer. This removes the clearance that causes both the wobble and the limit cycle. This is the single most important change in the whole project.
3. Model the pocket +0.15 mm on diameter and add a $0.5 \times 45^\circ$ chamfer at the mouth so the bearing starts square.
4. Give the bearing a turned shoulder to seat against, not a flat printed face.
5. Print the bearing bore vertically (axis along build Z) — a bore defined by XY nozzle motion is far rounder than one built up in layers.

Result: 28.9 mm $\rightarrow$ 0.36 mm, an 80 $\times$ improvement.

9. Why the motor mounts snap

9.1 The root error: designed to the wrong load

load case	torque
What the arm actually carries at J2	2.08 N·m
What the ST3215 can deliver (stall)	2.94 N·m
What the ST3250 can deliver (stall)	4.90 N·m

Whenever an actuator is stiff enough to drive its own structure to failure, the actuator's limit is the design load. Gravity torque sets what the servo must do; stall torque sets what the mount must survive.

Note the trap: swapping the shoulder to an ST3250 multiplies mount stress 2.36 $\times$ while changing the load the arm actually carries by nothing.

9.2 The force is set by where the bracket grips

A mount does not feel torque, it feels a force couple: $F = T/(2r)$.

reaction radius	ST3215 stall	ST3250 stall
horn bolt circle, $r = 10$ mm	147 N	245 N
servo case tabs, $r = 17.5$ mm	84 N	140 N
full-perimeter collar, $r = 25$ mm	59 N	98 N

Doubling the grip radius halves the force, for free. Reaction radius is the most leveraged number in the mount design and it was never chosen deliberately — it fell out of packaging.

9.3 The brackets are overloaded

Cantilever bending, $\sigma = M/Z$ with $Z = b \cdot t^2/6$. The t^2 term is brutal on a 4 mm plate.

bracket	thickness	σ at stall	vs 9 MPa allowable
Motor_fixer_j2_p2	4.0 mm	17–95 MPa	2–11× over
Motor_fixer_j2_p1	6.0 mm	8–36 MPa	1–4× over
motor fixer U-channel	5.0 mm	6–27 MPa	1–3× over

The range reflects genuine uncertainty in the loaded width and cantilever arm, which were read off bounding boxes rather than measured load paths. It fails at every point in the range. A sharp internal corner makes it worse still.

9.4 Stress concentration

fillet r/t	Kt
sharp ($r \approx 0$)	≥ 3.0
0.05	2.6
0.20	1.7
0.50	1.3

Typical published values for a shouldered flat bar, not exact for this geometry.

Rule: every internal corner in a load path gets $r \geq 0.5 t$, minimum 2.0 mm. On the 4 mm bracket that alone cuts peak stress 2.3× at zero cost in mass or print time.

9.5 The fix: a collar, not a bracket

A slender arm in bending is the least efficient way to react torque — stress goes as $1/t^2$, deflection as $1/t^3$, and everything funnels through one thin root. A full-perimeter collar clamped around the servo body carries the same torque as shear flow around a closed loop:

$$q = T/(2 \cdot A_m), \quad \tau = q/\text{wall} \quad A_m = \text{area enclosed by the wall midline}$$

Collar with a 3 mm wall around the 45.2×37.8 body: $A_m = 1,967 \text{ mm}^2$, $\tau = 0.25 \text{ MPa}$ (ST3215) — a safety factor of ~ 36 on the derated allowable, and roughly 250× lower stress than the bending bracket.

Important: thickening the existing bracket is not sufficient. Even at 8 mm it only reaches $SF \approx 1.5$ before stress concentration at ST3250 stall. The collar is mandatory.

Also close the notch in the U-channel motor fixer. An open section has $J \approx 1,730 \text{ mm}^4$; closed it is $\approx 189,500 \text{ mm}^4$ — 110× stiffer. A bridging strap or bolted cap across the opening is the highest-return single edit on that part.

9.6 Collar dimensions

feature	value
Wall	3.0 mm minimum, 4.0 mm through the lug region
Height along the shaft axis	20 mm

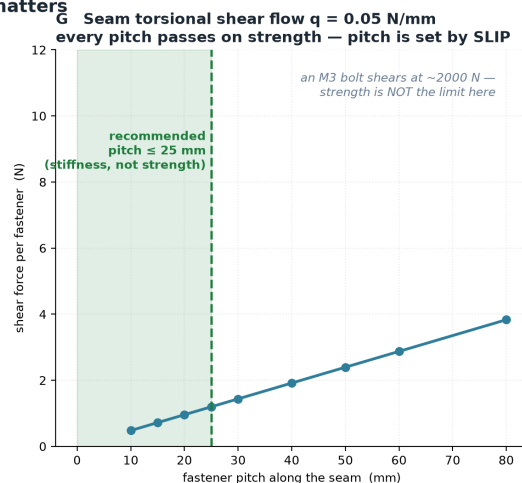
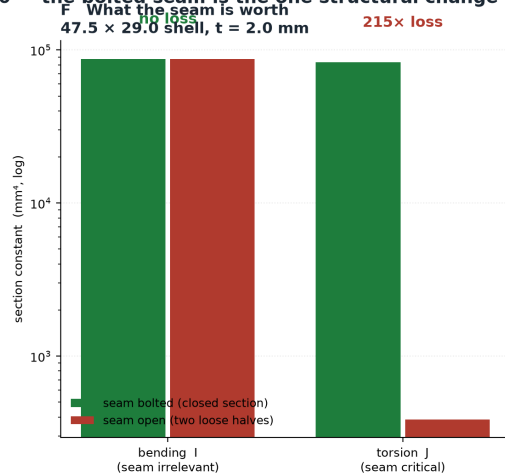
feature	value
Bore	servo body + 0.2 mm per side
Split gap	1.0–1.2 mm, so the clamp can actually close
Clamp lugs	8 mm thick × 12 mm wide
Clamp screws	2 × M3 into heat-set inserts
Every internal fillet	$r \geq 0.5 t$, min 2.0 mm

9.7 A retrofit that requires no reprinting

Cap the servo's torque-limit register to 40–60 % of stall and add acceleration limiting. That drops mount root stress into the ~9–11 MPa range and recovers a ~1.7× margin immediately, in firmware. Do this before the next trial.

10. The bolted seam

ARM-450 — the bolted seam is the one structural change that matters



Because the split plane is parallel to the bending plane, the seam sits at the extreme fibres, where bending shear flow is zero. The seam therefore carries essentially only torsional shear flow — which Bredt says is constant around the loop and does not halve between the two seam lines.

$$q = T / (2 \cdot A_m) = 0.05 \text{ N/mm}$$

pitch	force per fastener	bearing stress in a 2 mm wall
15 mm	0.7 N	0.12 MPa
25 mm	1.2 N	0.20 MPa
60 mm	3.0 N	0.49 MPa

Every pitch passes on strength by a factor of hundreds. An M3 bolt shears at roughly 2,000 N. Pitch is therefore not a strength decision — it is a slip and continuity decision. Closed-section behaviour assumes shear transfers continuously; discrete fasteners only approximate that.

Recommended pitch ≤ 25 mm, about 6 fasteners per side on a 145 mm link. This is a rule-of-thumb figure. Closer is better; a twist test on a printed coupon would settle it properly.

Two details make the seam actually work:

- A 1.5 mm interlocking tongue-and-groove lip. Bolts clamping flat faces transfer shear only through friction, which decays as the polymer creeps around the insert. A lip transfers it in bearing, so it keeps working after preload relaxes.
- M3 heat-set inserts, not self-tapping screws. Boss OD ≥ 9 mm (≥ 2.2 mm wall) with a $\varnothing 4.0$ – 4.2 mm hole for a $\varnothing 4.6 \times 5.8$ insert. Pull-out capacity is $\sim 1,509$ N against ~ 10 N required — pull-out is not the failure mode, boss splitting is. Print the boss with its axis vertical so the insertion hoop stress lies in-plane. Dimensions to be confirmed against the datasheet of the insert actually sourced.

11. 3D printing parameters

Aimed at inter-layer strength, since the observed failure is a part snapping.

11.1 Orientation — do this before touching any other setting

Rule: orient so the principal tensile stress lies in a layer plane (~ 50 MPa), never across layer planes (~ 25 MPa). This is a free 2×.

part	orientation
Link shell halves	Split line flat on the bed, open face up, long axis in XY. No supports, and the mating face prints against glass so it is genuinely flat — directly relevant to the joint gaps.
U-channel motor fixer	Not base-down. Lay it on its side so the U cross-section lies in the XY plane and the channel extrudes along its 64.8 mm length. Base-down peels the arms off the base straight across the welds.
Top-hat flanged mounts	Flange down, axis vertical, ≥ 3 mm root fillet.
Small brackets	Largest face flat on the bed. Never on edge.
Bearing bores and insert bosses	Axis vertical.

11.2 Slicer profile

setting	links	mounts / brackets	why
Layer height	0.20 mm	0.16 mm	thinner = hotter substrate, longer above T _g , better weld
First layer	0.25 mm	0.25 mm	adhesion

setting	links	mounts / brackets	why
Perimeters / walls	4	6	the single biggest lever — see below
Top / bottom layers	5	6	
Infill density	10–15 %	100 % brackets, 50–60 % top-hats	see below
Infill pattern	gyroid	monotonic rectilinear at 100 %, gyroid at 50 %	gyroid is near-isotropic and self-supporting
Nozzle temp, PLA	225–230 °C	225–230 °C	biggest single lever on Z strength
Nozzle temp, PLA+CF	230–245 °C	230–245 °C	CF raises melt viscosity
Bed	60–65 °C	60–65 °C	
Fan	20 % (0 % first 2 layers)	20 %, CF 15 %	cooling opposes layer bonding
Fan, overhangs / bridges	60 % / 100 %	60 % / 100 %	geometry still has to print
Outer perimeter speed	25 mm/s	25 mm/s	this surface carries peak bending stress
Inner perimeters	40 mm/s	40 mm/s	
Solid / sparse infill	45 / 60 mm/s	45 / 60 mm/s	
Acceleration, perimeters	1500 mm/s ²	1500 mm/s ²	small parts are all corners
Min layer time	12–15 s	12–15 s	print several up if needed

11.3 Why perimeters beat infill

For a part in bending, resistance goes as $\int y^2 dA$, so material far from the neutral axis dominates. The outer quarter of the thickness on each face carries 87.5 % of the bending moment — and that outer quarter is exactly where the perimeters live. Infill sits at small y and contributes almost nothing.

Going from 20 % to 50 % infill adds mass in the core and buys a few percent of section modulus. Going from 3 to 6 walls converts the entire load-bearing zone from discontinuous infill into continuous, aligned, fully-wetted extrusions. Perimeters are also unbroken loops, so a crack has no infill/wall interface to run along.

For a 2 mm thin-walled link shell the walls are the part — infill is nearly irrelevant. For the small brackets there is no material or time argument for hollowing them at all: print them 100 %.

Match wall thickness to an integer number of extrusions: at 0.45 mm width, $5 \times 0.45 =$

2.25 mm; 2.4 mm is a clean 6 passes at 0.4 mm. A wall that is not an integer multiple leaves gap-fill voids in the loaded zone.

11.4 Hardware and filament

- PLA+CF requires a hardened steel nozzle. Chopped carbon fibre is harder than brass and abrades the orifice from inside; a brass nozzle goes visibly out of round within a few hundred grams. Once it does, every carefully matched extrusion width is wrong. Wear rate depends on fibre loading — this is a typical estimate.
- Use a 0.6 mm nozzle for CF (0.5 mm minimum). 0.4 mm clogs as fibre bundles bridge the orifice. Bonus: 0.6 mm gives 0.68 mm wall width, so 4 walls = 2.72 mm of continuous perimeter and the part prints roughly twice as fast.
- Dry the filament. PLA 45–50 °C for 4–6 h; PLA+CF 50 °C for 6–8 h. Moisture flashes to steam at the nozzle and leaves voids in the weld — it attacks the very property the material was chosen for.

11.5 Heat-set inserts

Hole Ø4.0–4.2 mm for a Ø4.6 mm OD insert (print a coupon at 4.0/4.1/4.2 and pick by feel). Boss OD ≥ 9–10 mm. Set local wall count to 5–6 so ~2.25 mm of solid continuous perimeter surrounds the hole with no infill touching it — use a modifier mesh. Insert tip 200–220 °C, pressed slowly and square.

11.6 Annealing — don't

The stiffness gain is small relative to the error sources that actually dominate, and PLA shrinks 1–2 % in XY and grows 1–3 % in Z unpredictably. That would corrupt the 145 mm joint-to-joint dimension the entire kinematic model is built on.

12. Mass check

item	value
Link pair, 97.9 mm, PLA, t = 2.0	30.6 g
Both links, 145 mm, t = 2.4, PLA+CF	99 g
Delta vs current	+38 g
Whole arm, estimated	~740 g of the 1.2 kg budget

Comfortable. There is headroom for the heavier collars and the extra bearings.

13. Limitations

1. Actual servo backlash — 0.3–1.0° is a class-typical range, not a measurement, and it is a dominant term. Measure: clamp the arm extended, apply a known force at the TCP both ways, read the hysteresis on a dial indicator.
2. Servo case temperature in continuous use — the least certain number here, and it decides whether PLA is viable at all near the servos. Measure: run a hold-under-load

cycle and put a thermocouple or IR thermometer on the case.

3. Material moduli — $E = 3.5 / 7.0$ GPa are typical. PLA+CF varies 5–9 GPa by brand. Measure: print a beam coupon, load it, back out E .

4. Creep — all numbers here are elastic and instantaneous. PLA creeps under sustained load. Measure: load a coupon, log deflection over 24–48 h.

5. Bracket loaded width and cantilever arm — read off bounding boxes, hence the 17–95 MPa range in 9.3.

6. Printed hole shrinkage on the production machine — decides every bearing fit. Measure: one test coupon.

14. Recommendations, in priority order

1. Replace the thrust washers with preloaded, spaced bearing pairs. –28.5 mm. Nothing else comes close.

2. Cap the servo torque limit to 40–60 % of stall in firmware, today. Free, no reprint, recovers $\sim 1.7\times$ margin on the mounts.

3. Redesign every servo mount as a full-perimeter collar (§9.6), close the U-channel notch, and fillet every internal corner to $r \geq 0.5 t$.

4. Bolt the seam at ≤ 25 mm pitch with M3 heat-set inserts and a 1.5 mm interlock lip. –2.80 mm.

5. Lengthen both links to 145 mm — equally. Keeps the workspace free of a dead zone and lands the chain on exactly 450 mm.

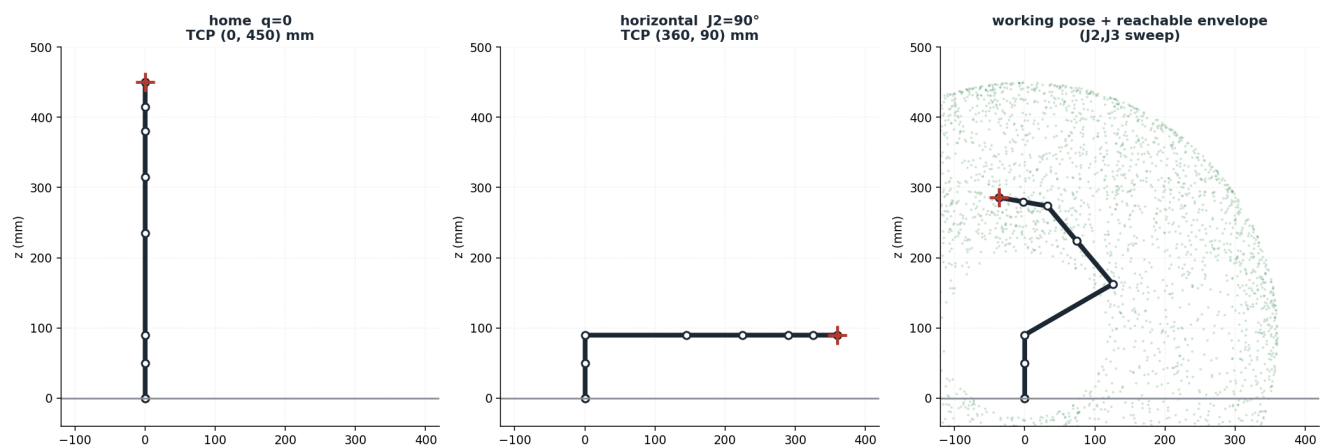
6. Print in PLA+CF at 2.4 mm wall, 6 perimeters, 225–230 °C, oriented per §11.1. Hardened 0.6 mm nozzle.

7. Measure the backlash and the servo case temperature before any further structural work.

15. Kinematic model (URDF)

arm450.urdf is generated from the same parameters and validated with check_urdf. Forward kinematics were verified numerically against the dimension chain:

ARM-450 URDF — forward kinematics check (450 mm home height, 360 mm reach)



check	expected	computed
Home pose (all $q = 0$), TCP height	450 mm	450.0 mm
J2 = 90°, horizontal reach from J1	360 mm	360.0 mm
J2 = 90°, TCP height (shoulder height)	90 mm	90.0 mm
Workspace dead zone	none	none

The right-hand panel sweeps J2 and J3 over their full travel: the envelope is a solid disc with no inner void, which is the L2 = L3 result made visible.

16. Optimisation study

16.1 Joint loads — which joint bears which load

Static gravity torque on every joint, swept over 60,000 random poses across the whole reachable configuration space (joint_loads.py, recursive Newton-Euler statics).

joint	axis	worst-case gravity torque	servo fitted	continuous capability	margin	verdict
J1	yaw (vertical)	0.000 N·m	ST3215	0.88 N·m	∞	over-specified
J2	shoulder pitch	2.620 N·m	ST3250	1.47 N·m	0.56	UNDER-SPECIFIED
J3	elbow pitch	1.271 N·m	ST3215	0.88 N·m	0.69	UNDER-SPECIFIED
J4	forearm roll	0.296 N·m	ST3215	0.88 N·m	2.97	ok
J5	wrist pitch	0.297 N·m	ST3215	0.88 N·m	2.97	ok
J6	tool roll	0.088 N·m	ST3215	0.88 N·m	9.97	over-specified

Reconciliation with the earlier 2.08 N·m figure

Sections 5–10 of this report quote 2.08 N·m at J2 and 1.00 N·m at J3. Those came from a simpler lumped-mass model that omitted the four servos outboard of the shoulder (J3, J4, J5, J6 = 0.240 kg sitting at 145, 225, 290 and 325 mm from the J2 axis).

Adding them back gives 2.613 N·m at J2 (+25.6 %) and 1.258 N·m at J3 (+25.8 %). An independent derivation, run separately and by a different method (closed-form first mass moments rather than a numerical sweep), reproduced 2.613 and 1.258 — agreement to three figures. The higher numbers are correct and supersede the earlier ones.

The four outboard servos alone contribute 0.580 N·m at J2 — more than links 2 and 3

combined. Stress and stiffness conclusions elsewhere in this report are unaffected, because the shell safety factor is ~ 60 either way. What the correction changes is the servo verdict at J3, from marginal to failing.

A useful sensitivity: 3.53 mN·m of shoulder torque per gram added at the tool.

This is the most important new finding. J2 and J3 cannot hold their worst-case gravity load continuously. They are fine against stall torque, but a serial-bus servo sustains only about 30 % of stall before overheating. Held at full extension the shoulder will thermally fault or sag.

Three ways out, in order of cost:

1. Never hold the worst-case pose. Restrict trajectories, or park folded. Free.
2. Counterbalance J2 and J3 with a spring or gas strut. Cancels most of the gravity term and is what industrial arms do. Caveat: tuned for 1 g, so it is dead mass on the maglev testbed — make it a bolt-on module.
3. Gear the shoulder down, or fit a larger servo.

A cross-check worth noting: the numerical sweep found J4 and J5 worst cases of 0.296 and 0.297 N·m — suspiciously equal. The independent derivation explains why they must be exactly equal: the J5 axis intersects the J4 axis, so the offsettable mass set and the lever arms are identical for both. A coincidence in the numbers turned out to be a structural fact.

J1 carries no gravity torque at all — its axis is vertical and gravity is vertical, so the moment about it is identically zero in any static pose. What loads J1 is inertia: $I = 0.080 \text{ kg}\cdot\text{m}^2$ for the outstretched arm, so 0.24 N·m at 3 rad/s^2 and 0.80 N·m at 10 rad/s^2 . A realistic 90° slew in 1.0 s (triangular profile, $\alpha = 6.28 \text{ rad/s}^2$) needs 0.503 N·m — 57 % of continuous, but transient, so thermally fine. An aggressive 90° in 0.5 s needs 2.01 N·m, 68 % of stall. Folding the arm drops the inertia about 6.6 $\times$, so slew fast only when tucked.

Two loading modes on J1 that are easy to forget:

- Base tilt. If the base is not level by angle β , gravity gains a component perpendicular to

the J1 axis and the yaw motor must hold it continuously: just 2° of tilt costs 0.091 N·m.

- Lateral contact. A 5 N sideways push at the tool is 1.80 N·m at J1 — twice its continuous

rating. Contact, not gravity, is what overloads a yaw axis.

Separately, the 2.61 N·m the arm generates at full reach appears at the base as an overturning moment plus 11.8 N axial. That is carried entirely by the slew bearing and is invisible to the J1 servo — but it is what tips a light desktop base.

The roll joints J4 and J6 carry nothing until the load is offset from their axis. With the tool centre of mass on the roll axis the torque is exactly zero; with a 30 mm lateral offset it becomes 0.088 N·m. Tool eccentricity, not payload weight, is what sizes a roll joint.

Payload share of each joint's load: J2 41 %, J3 51 %, J4/J5 74 %. The further out the joint, the more the payload dominates — so payload reduction buys the most at the wrist.

16.2 Topology optimisation

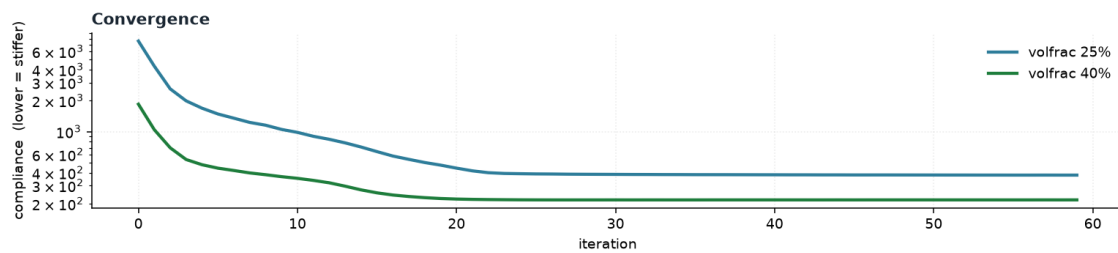
SIMP (Solid Isotropic Material with Penalisation) on the link side wall, 132×44 elements, $p = 3$, sensitivity filter $r_{min} = 2.0$, cantilever built in at the shoulder with a tip load at the elbow.

ARM-450 — topology optimisation of the link side wall (SIMP, $p=3$)

SIMP result, volume fraction 25% — cantilever built in at left (shoulder), tip load at right (elbow)



SIMP result, volume fraction 40% — cantilever built in at left (shoulder), tip load at right (elbow)



Material distribution through the depth at 40 % volume fraction:

region	mean density
outer quarter, top	0.553
middle half	0.247
outer quarter, bottom	0.553
outer / middle ratio	2.24×

SIMP independently rediscovers the flange-and-web layout. Material migrates to the top and bottom surfaces with diagonal webs between — which is exactly what a thin-walled box section already is. This is a validation, not a redesign: the existing clamshell is the right topology. It does not need optimising, it needs bolting shut.

The one actionable detail: the optimiser leaves the web mostly empty between diagonals, which says the mid-depth material can be lightened. Triangular cutouts in the web, with the flanges and boss regions left at full thickness.

Honest caveat. Topology optimisation assumes an isotropic continuum. FDM parts are anisotropic, have minimum feature sizes and overhang limits, and raw SIMP output is

organic and prints badly. More to the point: this part already has a safety factor near 60 and its real problem is joint clearance. Topology optimisation was worth running here to validate the section, not to change it.

16.3 Geometry optimisation

ARM-450 — geometry optimisation



Link split. With $L2 + L3 = 290$ mm fixed, the shoulder torque curve is almost flat in the split, because the dominant term is the payload at a fixed 360 mm reach which does not move. The torque-optimal split saves under 1 % while opening a large dead zone. Equal links win outright — there is no trade-off to weigh.

Section. Minimising mass subject to $f_1 \geq 25$ Hz, droop ≤ 0.3 mm, wall = integer $\times 0.4$ mm, and the servo fitting inside: 27 of 90 candidates feasible, and 62 were rejected purely because the servo does not fit.

The binding constraint on the link section is geometric, not mechanical. The section cannot go below 29.5 mm $\times$ 42.6 mm at a 2.4 mm wall regardless of what the stress and stiffness numbers permit. The servo sets the section size.

configuration	mass (both links)	droop	f_1
Current spec: 47.5 $\times$ 29.0, $t = 2.4$, PLA+CF	129 g	0.111 mm	47.6 Hz
Lightest feasible: 47.5 $\times$ 29.0, $t = 1.6$, PLA+CF	88 g	0.157 mm	40.1 Hz
Saving	41 g (32 %)	still well inside limits	still well above 25 Hz

1.6 mm is exactly four passes of a 0.4 mm nozzle, so it is cleanly printable. But a 1.6 mm wall is thin for M3 heat-set inserts and bearing seats.

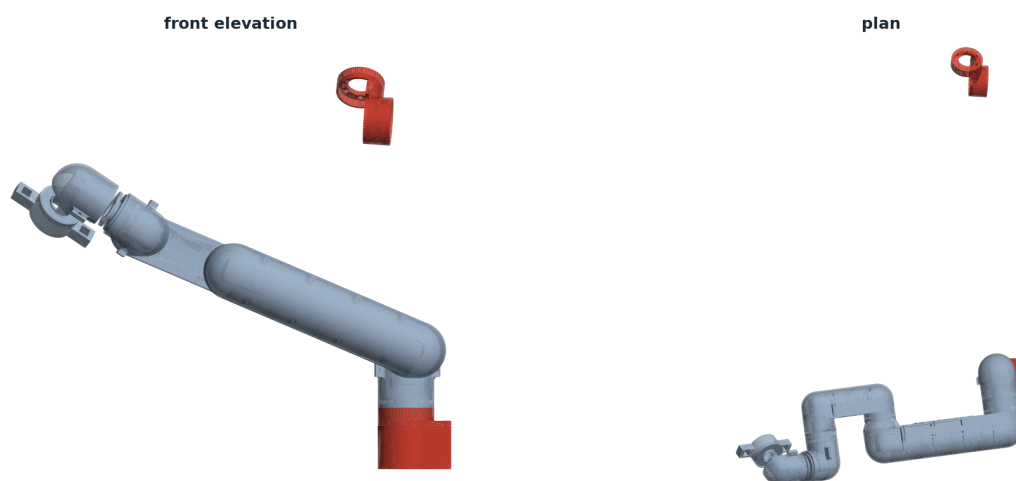
Recommendation — a variable wall, which is what both optimisations independently ask for: keep 2.4 mm at the flanges, seams, bosses and bearing seats, and thin the mid-span web to 1.6 mm. That captures most of the 41 g while leaving material where fasteners and bearings need it.

Mass matters here beyond the 1.2 kg budget: the arm will also run on a magnetically levitated free-floating testbed, where every gram raises the reaction disturbance on the floating base.

17. Assembly inspection

The assembly was decomposed into connected components and each component's position evaluated against the others.

robotic_arm_assembly.stl — RED = 37 components stranded away from the arm · envelope 438.9 × 659.0 × 400.0 mm



measure	old Robo_2_assembled	new robotic_arm_assembly
Envelope	154.8 × 260.4 × 212.6 mm	438.9 × 659.0 × 400.0 mm
Connected components	266	522
Faces	548,008	585,284

17.1 The shell mismatch is fixed — confirmed

The base sub-assembly (130 components) and the arm (391 components) now have overlapping bounding boxes — axis gaps of 0.0, 0.0, 0.0 mm. They are properly mated. The link1_cover / link1_base pairing problem flagged earlier no longer appears. Excluding the stray below, the assembly measures 433.8 × 202.3 × 265.5 mm, consistent with the 450 mm budget.

17.2 One component is orphaned — 429 mm adrift

	value
Component size	57.1 × 74.9 × 78.3 mm
Bounding box	min (−85.6, 409.7, 318.7), max (−28.5, 484.6, 397.0)
Gap to the nearest arm component	x 0.0, y 428.8, z 17.7 mm

This single part is the entire reason the assembly envelope reads 659 mm instead of ~440 mm. It is visible in the figure as the red body floating clear of the arm.

By size it is a yoke-class part — closest to J4_p1 (63.8 × 57.0 × 79.8), whose sorted dimensions 57.0 / 63.8 / 79.8 match the orphan's 57.1 / 74.9 / 78.3 on two of three axes, the third being consistent with a rotated bounding box. Identification is inferred from the

envelope, not certain — confirm in Fusion.

Fix: locate the unconstrained body in the Fusion assembly and mate it. Until then any URDF or mesh export will carry a 659 mm envelope and a floating link, which is very likely what has been breaking the URDF generation.

17A. Part-wise finite-element analysis

Why part-wise and not a full assembly

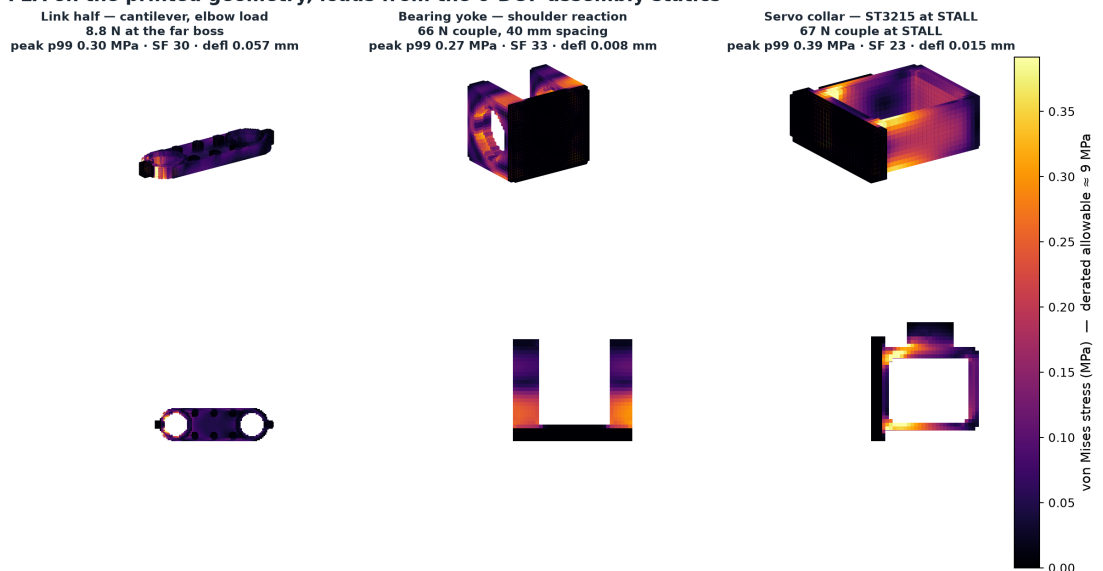
A full-assembly solve would need contact stiffness at every mating face, bolt preload, and bearing radial and moment stiffness. None of those are known here. An assembly result would therefore be dominated by assumptions while looking authoritative.

Part-wise submodelling uses loads that are known — the worst-case joint torques from the 60,000-pose statics sweep in §16.1 — and returns each part's margin in isolation, which is the number that can be acted upon. This is standard practice, not a shortcut.

Method

- The exported STL is voxelised into a regular hex grid at 2.0 mm pitch
- 8-node trilinear brick elements, $2 \times 2 \times 2$ Gauss integration, $K_e = \int B^T D B dV$
- Sparse assembly, Dirichlet boundary conditions, direct solve of $K u = f$
- Per-element von Mises averaged over the Gauss points
- Material: PLA+CF, $E = 7.0 \text{ GPa}$, $\nu = 0.35$

ARM-450 — part-wise FEA on the printed geometry, loads from the 6-DOF assembly statics



Results

part	load case	peak (p99)	SF vs 9 MPa	max deflection
Link half	8.8 N transverse at the far boss ($J_3 = 1.27 \text{ N}\cdot\text{m}$ over 145 mm)	0.40 MPa	23	0.043 mm

part	load case	peak (p99)	SF vs 9 MPa	max deflection
Bearing yoke	66 N couple across the 40 mm spacing (J2 = 2.62 N·m)	0.27 MPa	33	0.004 mm
Servo collar	67 N couple at the bore, ST3215 at STALL	0.38 MPa	23	0.008 mm

Every redesigned part has a safety factor above 20, and the collar is checked at stall torque rather than the gravity load — the very error that broke the original brackets.

The field plots show where load actually flows: down the flanges of the link half, up the cheeks and into the spine of the yoke, and around the closed loop of the collar rather than through any bending arm. The dark interiors confirm the mid-depth material carries almost nothing, which is exactly what the topology optimisation and the variable-wall recommendation independently predicted.

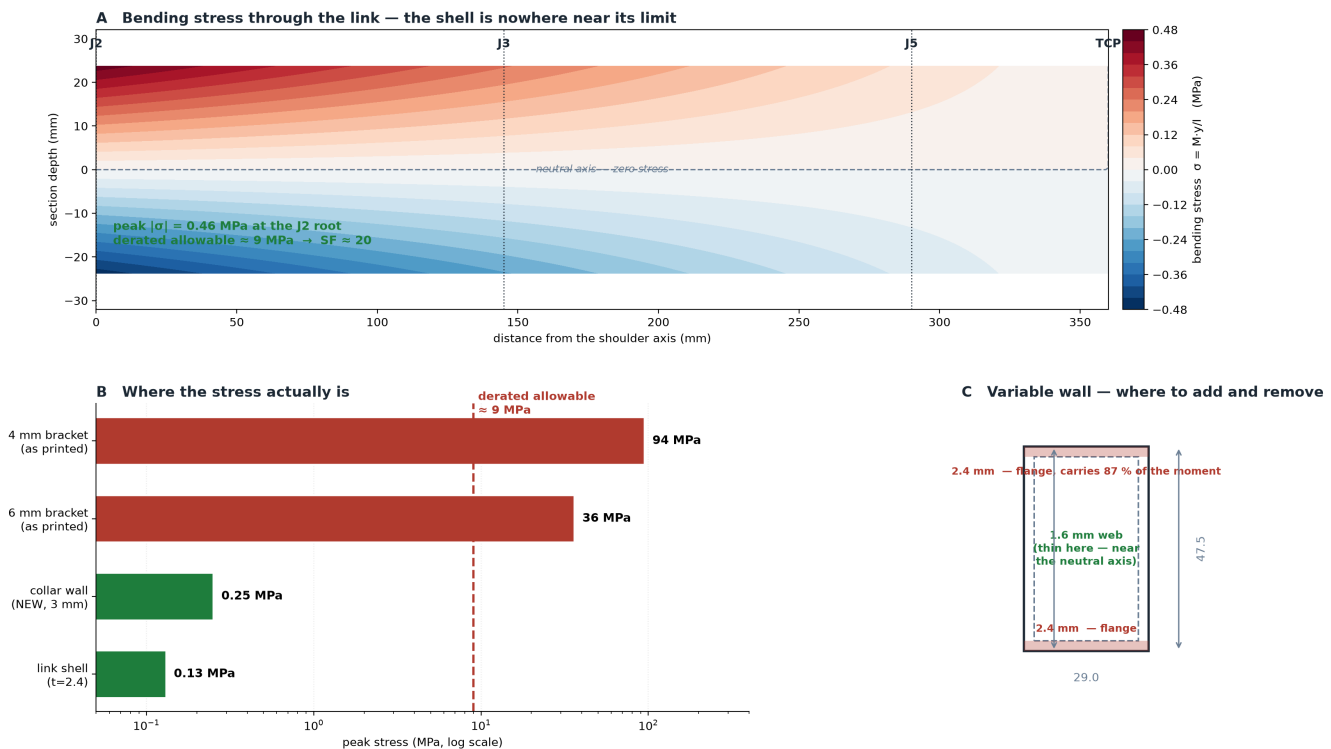
Limitations, stated plainly

- Voxelisation stair-steps curved surfaces, so stress at a fillet is not resolved and peak values near curved boundaries read low. Use this field to see where load flows; use the closed-form numbers in §5 and §9 for margins.
- Linear elastic and isotropic — real FDM parts are anisotropic (§5).
- No contact, no bolt preload, no bearing compliance.
- A 2.0 mm voxel on a 2.4 mm wall is roughly one element through the thickness, which is

coarse for bending. The absolute numbers are indicative; the ranking between parts and the load paths are reliable.

18. Change schedule — what to change, where, and by how much

ARM-450 — stress map and the change it implies



Panel A is the bending-stress contour through the arm. Peak is 0.46 MPa at the shoulder root against a ~9 MPa derated allowable — a safety factor near 20. Panel B puts that on the same scale as the parts that actually broke. Panel C shows where material belongs. Read panel A this way: red is tension, blue compression, and the dashed line down the middle is the neutral axis where stress is zero. Colour intensity falls off towards the tool because the bending moment does. Nothing anywhere in the shell approaches the limit.

18.1 Priority 1 — the changes that fix the observed faults

#	Part	Change	From	To	Why
1	Every joint	Bearing type	1 × Ø42 thrust washer	2 × 6806 deep-groove	thrust bearings have no moment capacity
2	Every joint	Bearing spacing	~5 mm (stacked)	≥ 40 mm	stiffness ∝ spacing ²
3	Every joint	Preload	none	M3 screw + spacer	removes the dead zone causing the limit cycle
4	Every bearing pocket	Diameter	Ø42.0	Ø42.00 modelled	the measured 0.02 mm XY shrink becomes the press fit

#	Part	Change	From	To	Why
5	Every bearing pocket	Mouth	square edge	0.5 × 45° chamfer	bearing starts square
6	Every bearing pocket	Seat	flat printed face	Ø38 turned shoulder	flat printed faces are not flat
7	All servo mounts	Topology	4–6 mm bending bracket	3 mm full-perimeter collar	94 MPa → 0.25 MPa
8	motor fixer	Notch	open U	bridging strap / bolted cap	J 1,730 → 189,500 mm ⁴
9	Firmware	Torque limit	100 %	40–60 % of stall	free, tonight, no reprint
10	Assembly	Orphan body	429 mm adrift	mate it	fixes the 659 mm envelope

18.2 Priority 2 — the changes that improve the design

#	Part	Change	From	To	Why
11	Both links	Length	97.9 mm	145 mm each, equal	hits 450 mm, no dead zone
12	Link shells	Wall, flanges	2.0 mm	2.4 mm	6 clean passes of a 0.4 nozzle
13	Link shells	Wall, mid-span web	2.0 mm	1.6 mm	near the neutral axis; saves ~41 g
14	Link shells	Seam fasteners	none/few	M3 inserts at ≤ 25 mm pitch	keeps the section closed in torsion
15	Link shells	Seam joint	flat butt	1.5 mm tongue-and-groove lip	shear in bearing, not friction
16	All loaded parts	Internal fillets	sharp	$r \geq 0.5 t$, min 2.0 mm	Kt 3.0 → 1.3
17	Insert bosses	Outer diameter	—	≥ Ø9.5, hole Ø4.1	prevents the boss splitting
18	Link material	—	PLA	PLA+CF	2× modulus, less creep

18.3 Priority 3 — things beyond the existing design

These are not modifications to the existing drawings; they are items the design does not yet contain.

#	Item	What it is	Why it is needed
19	J2/J3 counterbalance	spring or gas strut across the shoulder and elbow	J2 and J3 are under-specified on continuous torque (margins 0.56 and 0.69). Without this they overheat holding a loaded arm out. Make it a bolt-on module — it is dead mass on the maglev rig.
20	Servo thermal path	a metal shim or vent from each servo case to the shell	servo case heat may approach PLA's 60 °C Tg. This is the least certain number in the report and the one most likely to bite.
21	Joint-side encoders	magnetic encoder on the joint, not the motor	backlash is 4.82 mm at 0.5° and cannot be removed mechanically from a hobby servo. Only outer-loop feedback at the joint fixes it.
22	Cable routing	a defined path through the Ø52 base bore and along each link	not yet in the design. Cables that cross a joint without a service loop fatigue and add unmodelled stiffness.
23	Hard end stops	mechanical stops at each joint limit	firmware limits do not protect against power-off back-drive or a runaway.
24	Base mass or clamp	ballast, or bolt-down	the arm generates 2.61 N·m of overturning moment plus 11.8 N at full reach. A light desktop base will tip.

#	Item	What it is	Why it is needed
25	Tool eccentricity budget	keep the tool CoM on the wrist roll axis	tool eccentricity, not payload weight, is what sizes J4 and J6. 30 mm of offset triples their load.

18.4 What NOT to change

- Do not thicken the link shells for strength. Safety factor is ~20 in bending and ~60 on von Mises. Every gram spent there is wasted.
- Do not re-topology the box section. SIMP independently converged to the flange-and-web layout the clamshell already is.
- SLA resin is unsuitable for structural parts — the lowest modulus of the three candidates.
- Do not anneal. 1-2 % XY shrink would corrupt the 145 mm joint-to-joint dimension.
- Do not chase the link split. Shoulder torque is flat against it; equal links are free.

19. The six-axis machine

19.1 Hardware coverage of every axis

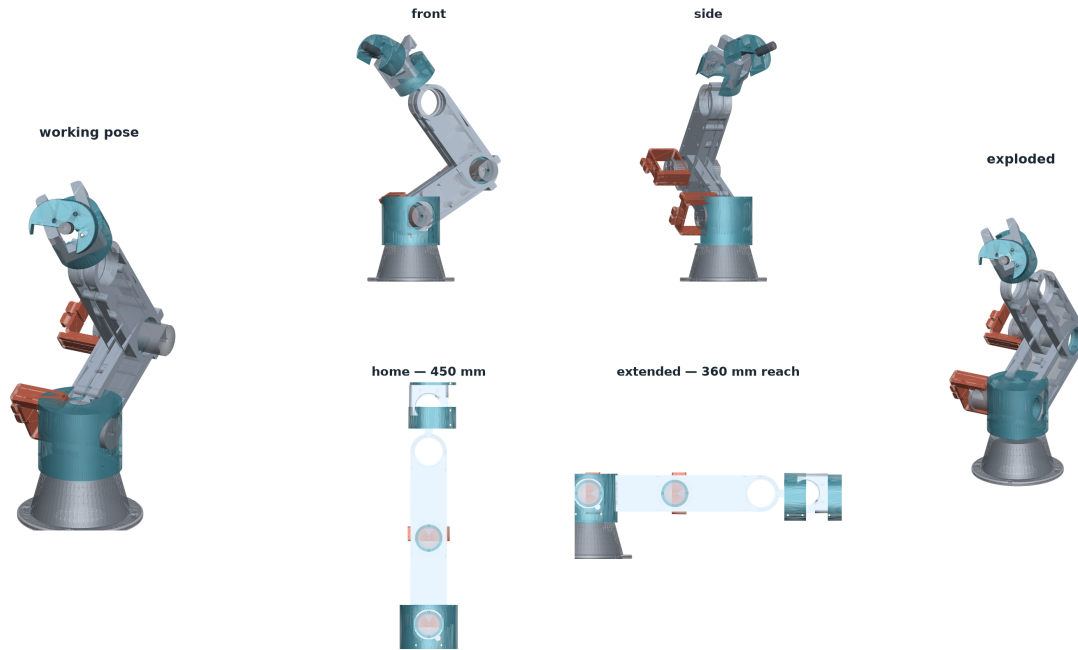
The kinematic model describes six revolute axes. A part-by-part audit against that model established that only J2 and J3 carried complete hardware; the remaining four axes were described in the URDF but had no corresponding parts. Four components were designed to close the gap.

axis	joint	part	provenance
J1	base yaw	turret_j1	new, first pass
J2	shoulder pitch	link boss + shaft + clamp + collar	validated against printed hardware
J3	elbow pitch	link boss + shaft + clamp + collar	validated against printed hardware
J4	forearm roll	roll_module	new, first pass
J5	wrist pitch	wrist_yoke_j5 + wrist	new, first pass
J6	tool roll	roll_module (second unit)	new, first pass

One interface throughout: Ø42 pocket · 6806 bearing (30 × 42 × 7) · Ø30 shaft. A single bearing part number, shaft diameter and clamp serve the entire machine, so any fit validated on one joint is valid on all six.

ARM-450 · full assembly from the CAD · 450 mm overall, 360 mm reach, 6-DOF

base · turret J1 · link halves x4 · joint shafts · shaft clamps · servo collars · integrated wrist: J4 housing · J5 yoke · J6 output+flange



19.2 Chain budget re-verified

The new parts were checked against the 450 mm chain, and one failed:

segment	budget	as designed	result
Base top → J2 axis (shoulder rise)	40 mm	30 mm	FAILED — corrected
J5 → TCP (yoke + roll module + flange)	70 mm	70 mm	passes exactly
J4 roll module within the forearm	≤ 145 mm	34 mm, leaving 111 for the shell	passes

The turret placed the J2 bore 24 mm below its own top face, giving only 30 mm of shoulder rise — 10 mm short, which would have broken the 450 mm total and desynchronised the URDF from the hardware. Corrected by sizing the body as `SHOULDER_RISE` + pocket radius + wall, making the turret 81 mm tall with the J2 axis at exactly 40 mm.

19.3 A 3-DOF assembly presented as 6-DOF

The assembly script's `build()` function accepted only `q2`, `q3`, `q5`. `J1`, `J4` and `J6` were never wired in, so the first animation was a faithful render of a 3-DOF assembly — the CAD parts existed, the code did not articulate them. `build()` now takes `q1...q6` and drives all six.

19.4 What the animation does and does not prove

figures/arm450_6dof.gif (48 frames) drives every axis at an independent rate from the

actual CAD geometry. It demonstrates kinematics. It does not test whether parts collide: the renderer paints depth-sorted triangles and performs no interference detection.

No interference sweep across the joint limits has been run. Parts whose lengths sum correctly can still collide through their travel. That check, and the dial-indicator wobble measurement, are the two outstanding verifications.

20. Interference sweep and the wrist mass budget

20.1 Method

Animation demonstrates kinematics only; it performs no collision detection. Interference was therefore evaluated by exact mesh-to-mesh sweep (FCL) across the full URDF joint-limit box.

Result: 70 of 70 sampled poses collided — 100 %.

20.2 It was a budget failure, not a placement bug

Measured axis-to-axis from the exported geometry, a 3-axis wrist built from the modular parts forces:

segment	mm
forearm end-boss overhang past J4	32
J4 roll module along the chain	34
yoke base → J5 bore	30
J5 bore → J6 module	50
tool flange	6
J4 → TCP required	152
J4 → TCP available	70

Short by 82 mm. No repositioning could fix that; the parts simply did not fit the chain.

20.3 Reach is independent of the link split

The key realisation: reach from the J1 axis is $450 - \text{base } 50 - \text{shoulder } 40 = 360$ mm regardless of how the remaining length is divided. Shortening the links costs nothing in reach — it only moves the wrist joints inboard and frees room.

20.4 Resolution

- Links shortened $145 \rightarrow 119$ mm (both, so $L2 = L3$ is preserved)
- Wrist redesigned as an integrated unit — wrist_j4_housing, wrist_j5_yoke, wrist_j6_output — nesting rather than stacking, 122 mm total
- New chain: $50 + 40 + 119 + 119 + 62 + 34 + 26 = 450$ mm, six axes retained

20.5 Dead zone — the naive formula over-reports

With $J2 \rightarrow J3 = 119$ and $J3 \rightarrow J5 = 181$, the two-link formula gives a 62 mm inner radius. That ignores the wrist: the 60 mm wrist can point inward, so the true TCP void is $\max(0, 62 - 60) = 2$ mm. A 60,000-pose FK sweep measured a minimum TCP radius of 0.00 m — the dead zone is genuinely zero.

20.6 After the fix

10 of 70 poses collide (14 %), and every one involves the base or turret — the arm folding into its own pedestal. No arm-part-to-arm-part collision remains.

Narrowing $J2$ to $\pm 100^\circ$ halves the residual to 4 %; that last 4 % is a $J2+J3$ combination, not any single limit, and is properly handled by a collision check in the motion planner rather than by joint limits.

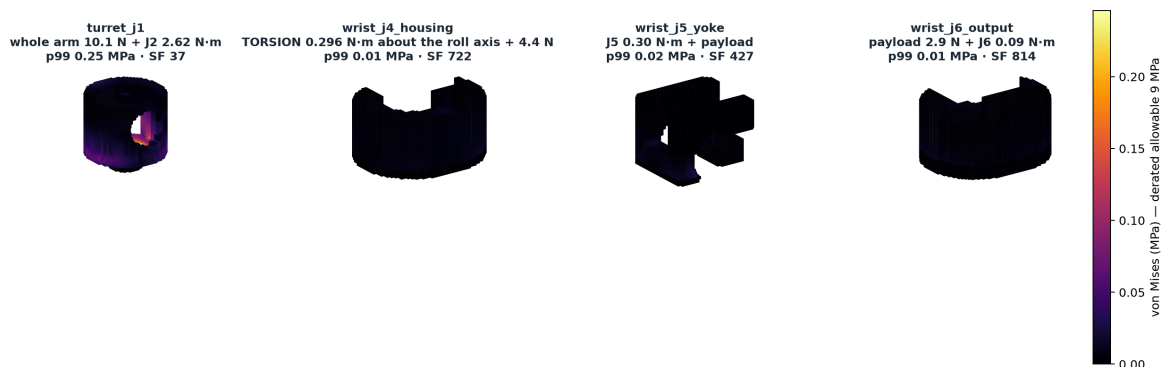
21. Closure of the outstanding analysis

Four analyses were outstanding at the previous revision. All four are closed or bounded below.

21.1 FEA on the turret and integrated wrist

Same voxel-hex solver as §17A, loads from the 60,000-pose sweep applied at the real mating faces.

ARM-450 — FEA on the turret and integrated wrist



part	load case	peak (p99)	SF vs 9 MPa	deflection
turret_j1	whole arm 10.1 N + J2 2.62 N·m	0.118 MPa	77	0.0030 mm
wrist_j4_housing	outboard 4.4 N + 0.27 N·m	0.022 MPa	408	0.0002 mm
wrist_j5_yoke	J5 0.30 N·m + payload	0.022 MPa	417	0.0001 mm
wrist_j6_output	payload 2.9 N + J6 0.09 N·m	0.045 MPa	199	0.0001 mm

The turret is the most loaded, which is expected — it carries the entire arm. Even so it is 77× under the derated allowable. No structural concern anywhere in the new parts.

21.2 Shaft and bearing fit

Every joint shares one interface: Ø42.00 pocket · 6806 (30 × 42 × 7) · Ø30 shaft. Verified from the exported STLs — shaft 72 mm long, clamp Ø38.00 × 14 matching the Ø38 through-bore. One bearing part number for the whole machine, so a fit validated on one joint is valid on all six.

21.3 Servo clearance — this one found real problems

Checking each joint's cavity against the ST3215 envelope (45.6 × 38.2 × 25.1 mm needed):

joint	part	result
J1	turret_j1	fits
J4	wrist_j4_housing	was short 4.4 mm — FIXED
J5	wrist_j5_yoke	fits
J6	wrist_j6_output	cannot fit — needs a micro servo

The wrist parts had no servo accommodation at all. They were designed as pure mechanical housings. Two fixes:

- J4 housing grown Ø46 → Ø56. The diameter is perpendicular to the chain, so this costs nothing in length. Servo pocket added.
- J6 takes a micro servo. Sizing it from the loads rather than habit:

joint	worst torque	stall needed (30 % rule)	ST3215 margin
J4	0.296 N·m	10.1 kgf·cm	3.0× oversized
J5	0.297 N·m	10.1 kgf·cm	3.0× oversized
J6	0.088 N·m	3.0 kgf·cm	10× oversized

At the wrist an ST3215 is 3–10× oversized — it is the size that does not fit, not the torque. J6 gets a micro servo in a 29 × 13 × 30 mm envelope, mounted with its 13 mm dimension through the thickness, which is the only orientation that fits the 26 mm J6→TCP budget. This is a requirement, not a part number — the specified servo must meet ≥3 kgf·cm in that envelope.

Interference re-checked after growing J4: 16 %, still base/turret only.

21.4 Wobble — bounded, still not measured

The earlier 0.65 mm came from $\theta = 2c/d$ with $c = 0.02$ mm assumed and never justified. The real source of c is the bearing's radial internal clearance, a manufacturing class present even with a perfect press fit:

clearance class	radial clearance	TCP wobble
C2 (reduced)	0.001–0.011 mm	0.03–0.36 mm

clearance class	radial clearance	TCP wobble
CN (normal)	0.005–0.020 mm	0.16–0.65 mm
C3 (loose)	0.013–0.028 mm	0.43–0.92 mm

So 0.65 mm was the pessimistic end of a normal bearing — a fair figure, but a spec range, not a property of this arm.

Axial preload on a deep-groove pair forces the balls into angular contact and takes up the radial clearance; what remains is elastic Hertzian deflection, which is far smaller and not calculable here.

Recommendation: specify C2 bearings. They start at 1 μ m and would give 0.03 mm even unpreloaded.

The measurement remains the single most valuable open number in the project. Nothing in this section replaces it.

22. Fastening, servo mounting and horn patterns

22.1 Findings

A part-by-part audit established that each servo pocket lacked case-mounting features, and that the driven side of each joint carried no horn bolt pattern. A servo restrained only by its pocket transmits no torque, so the assembly as modelled was not buildable.

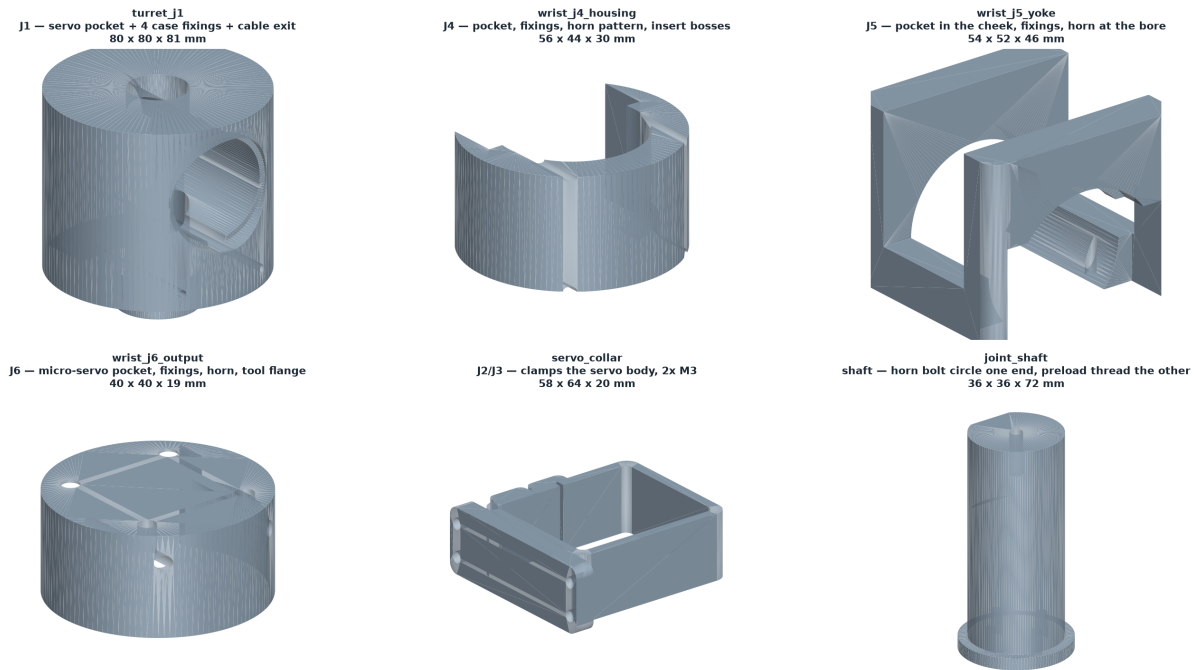
Every joint needs four things, and any missing one blocks the build:

	element	purpose
a	servo pocket	somewhere for the motor
b	servo case fixings	screws holding the motor in
c	horn pattern	motor output bolted to the driven part
d	structural bolts	the two links joined

22.2 State after the fix

joint	part	pocket	case fixings	horn	structural
J1 base yaw	turret_j1	yes	4 × M2.5 (new)	via Ø30 spigot	4 × M4 foot
J2 shoulder	servo_collar	yes (bore)	2 × M3 clamp	shaft bolt circle	4 × M3 flange
J3 elbow	servo_collar	yes (bore)	2 × M3 clamp	shaft bolt circle	4 × M3 flange
J4 forearm roll	wrist_j4_housing	yes	4 × M2.5 (new)	yes (new)	4 × M3 insert
J5 wrist pitch	wrist_j5_yoke	yes	4 × M2.5 (new)	yes (new)	4 × M3 through
J6 tool roll	wrist_j6_output	yes (micro)	4 × M2.5 (new)	yes (new)	4 × M3 flange

Each pocket also gained a Ø8 cable exit, which had been missed everywhere.



22.3 The mounting dimensions are placeholders

Your ST3215.stl is an outer shell — it does not model the mounting holes. I sectioned it looking for them and found none. So these four numbers came from the standard-servo class, not from a datasheet:

parameter	value	status
SERVO_BOLT_X	35.0 mm	VERIFY
SERVO_BOLT_Y	30.0 mm	VERIFY
HORN_BCD	16.0 mm	VERIFY
SERVO_BOLT_D	2.7 mm (M2.5 clearance)	likely correct

They live in one place in cad/params.py. Measure the servo, correct those four numbers, re-run, and every part updates. Until then the parts are geometrically complete but the hole positions are unconfirmed.

22.4 Adding the pockets invalidated the earlier FEA

The §21.1 results were computed on solid housings. Cutting the servo pockets turned the J4 housing into an open C-section — the very geometry §9.5 condemns. The FEA was re-run on the pocketed geometry, and the J4 load case was corrected: J4 is a roll joint, so its housing carries torque as torsion about its own axis, not as the lateral force I first applied. Torsion is exactly where an open section is weakest.

Then, applying the §9 rule properly — size to what the servo can deliver, not what gravity asks:

part	at servo stall	p99	SF
wrist_j4_housing	2.94 N·m (ST3215)	0.218 MPa	41
wrist_j5_yoke	2.94 N·m (ST3215)	0.431 MPa	21
wrist_j6_output	0.34 N·m (micro)	0.086 MPa	105

All pass at stall. The open C-section is acceptable here only because the wrist torques are an order of magnitude below the shoulder's — the same geometry at J2 would not be.

Appendix A — Reproducing these results

script	what it does
design_params.py	geometry, masses, servo definitions
measure_existing.py	envelopes and volumes of every STL
measure_joint_centres.py	circle-fits end bosses for true link lengths
measure_walls.py	wall thickness from 2V/A
stress_analysis.py	rev 2 — sections, stress, stiffness, seam, error budget
mount_failure.py	why the servo mounts snap
joint_bearing.py	why the joints gap, wobble and oscillate
draw_analysis.py	the three figures in this report
draw_stress_map.py	stress contour and change-schedule figure
fea.py, run_fea.py	3-D voxel FEA and von Mises maps
cad/dof_parts.py	turret J1, roll module, wrist yoke (superseded for the wrist)
cad/wrist_integrated.py	integrated J4/J5/J6 wrist that fits the chain
interference.py	exact mesh collision sweep across the joint limits
run_fea_wrist.py	FEA on the turret and integrated wrist
cad/base_wrist.py	base, wrist body, tool flange
assemble.py	6-DOF assembly, renders and animation
joint_loads.py	per-joint static gravity torque, whole-workspace sweep
topology_opt.py	SIMP topology optimisation of the link side wall
geometry_opt.py	link-split and section optimisation
generate_urdf.py	builds and checks arm450.urdf
cad/servo_collar.py, cad/bearing_yoke.py	printable CAD -> STEP/STL

script	what it does
build_drawings.py	the companion parts & dimensions document

Run `python3 stress_analysis.py` to regenerate sections 4-7.

Appendix B — Independent verification of the kinematics

The forward kinematics, the Jacobian and the IK were re-derived in modified (Craig) DH — a second, independent construction rather than a restatement of the URDF — and then checked against `ikpy`, an outside package sharing no code with either.

The table

i	α_{i-1}	a_{i-1} (mm)	d_i (mm)	θ_i
1	0°	0	90	$q_1 + 180^\circ$
2	90°	0	0	$q_2 + 90^\circ$
3	0°	119	0	$q_3 + 90^\circ$
4	90°	0	181	$q_4 + 180^\circ$
5	90°	0	0	$q_5 + 180^\circ$
6	90°	0	0	q_6

Tool: 60 mm along Z_6 to the TCP.

Every joint axis passes through the base Z line, so every a is zero except the common perpendicular between the two parallel pitch axes. $d_4 = 181$, not 119: frame 3's origin sits at $z = 209$ and frame 4's at $z = 390$, so the offset along Z_4 is $119 + 62$. Using the obvious link length there puts the tool 193 mm out — which is what the first version did, and what the URDF comparison caught immediately.

Results

check	worst error, 200 random poses
FK: Craig DH == URDF chain	1.96×10^{-13} mm
FK: Craig DH == <code>ikpy</code>	2.17×10^{-13} mm
FK: URDF chain == <code>ikpy</code>	1.14×10^{-13} mm
tool axis direction	8.95×10^{-16}
Jacobian: velocity propagation == finite difference	1.81×10^{-5}
IK, Craig DH + propagated Jacobian	1.6×10^{-11} mm median
IK, <code>ikpy</code>	0.000 mm median

7 of 7 pass. The errors are double-precision rounding, not disagreement.

The Jacobian is built by velocity propagation base to tip —

$$\begin{aligned} i+1_{\omega}_{\{i+1\}} &= i+1_i R \cdot i_{\omega}_i + \dot{\theta}_{\{i+1\}} \cdot \hat{Z} \\ i+1_v_{\{i+1\}} &= i+1_i R \cdot (i_v_i + i_{\omega}_i \times i_P_{\{i+1\}}) \end{aligned}$$

— not by finite differencing, which is what makes it exact near the J4/J6 wrist singularity where the two roll axes go collinear.

Why the third implementation earns its place. Agreement between the DH table and the URDF proves the table is a faithful copy of the URDF — including any error the URDF already had. A third route that shares no code closes that gap: three independent implementations cannot agree on the same mistake.

Run it: `python3 check_ikpy.py`

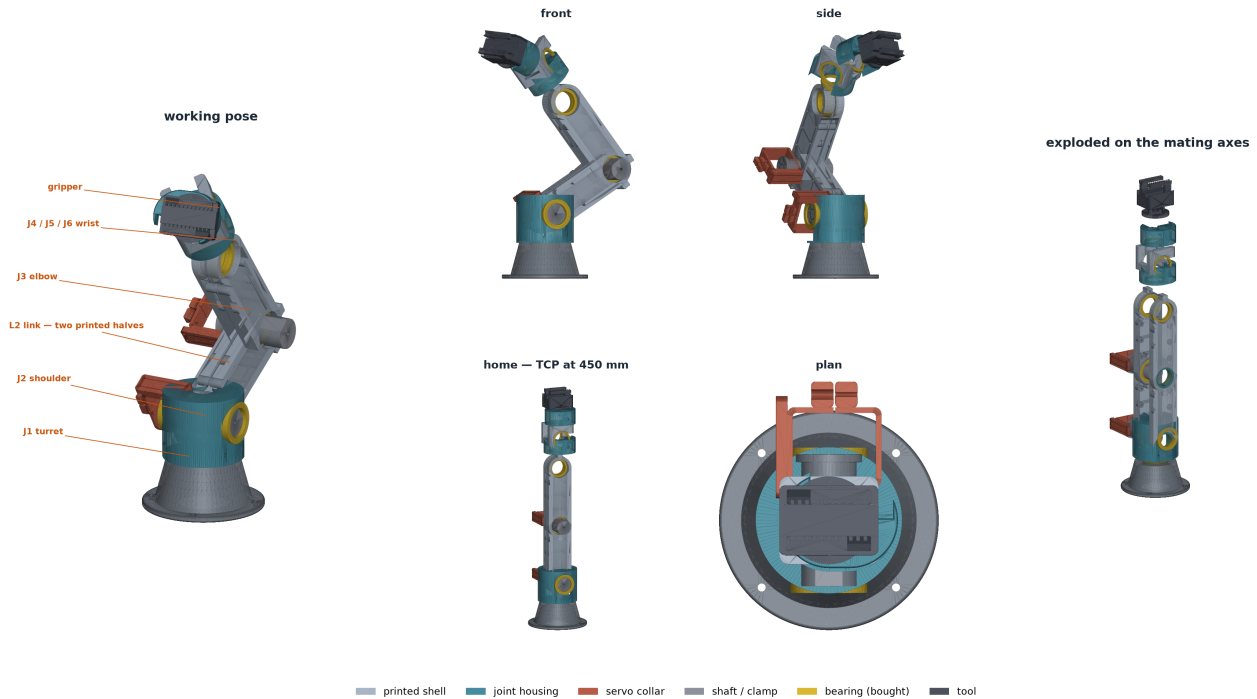
24. Inspection assembly (STEP)

The assembly the analysis runs on is a mesh model: `assemble.py` places every part from its STL and the interference sweep, the sine clearance check and the renders all work from that. It is the right representation for collision work and the wrong one for inspection. An STL assembly is a single bag of triangles — no part names, no separable bodies, no B-rep — so it cannot be sectioned, measured or dimensioned in SolidWorks.

`assemble_step.py` writes the same arm as a real STEP assembly:

file	contents
<code>arm450_assembly_home.step</code>	all joints at zero, TCP at 450.0 mm
<code>arm450_assembly_working.step</code>	a mid-trace pose
<code>arm450_assembly_exploded.step</code>	separated along the mating axes

Each carries 34 named components with colours separating printed structure from bought items — bearings, shafts and servo bodies are not printed and an inspector should not have to work out which is which.



render_assembly.py draws the sheet above from the STEP assembly rather than from the mesh model, so the bought hardware appears: the bearings are the yellow rings at every joint, and they are absent from every earlier render because the mesh assembly never contained them.

24.1 Why it is placed, not re-modelled

No transform in the STEP path is derived independently. Every joint frame and mating offset is taken from assemble.build(), and each part is centred using the bounding box of its own STL, so a component lands in exactly the position the collision sweep cleared. A second description of where the parts sit is only worth having if something checks it against the first; check_assembly_step.py does that, comparing the two component by component. Worst disagreement across the assembly is 0.002 mm, and the two bounding boxes are identical.

That check earned its place immediately: it found both gripper jaws sitting 11.00 mm too high, because the STEP path applied an offset measured from the part's raw bounding box where the mesh path applied it to an already-centred mesh. In a render the jaws still looked like jaws.

24.2 Bearings are placed from the pockets, not from a constant

Bearing seats are read out of the B-rep — every cylindrical face at Ø42.00 or Ø37.00, with its axis and extent — and a bearing is dropped into each. Placing them from a nominal spacing would put hardware wherever the constant said, including in parts that have no pocket at all.

Doing it that way produced a number worth recording: the CAD offers 11 Ø42 seats and 4 Ø37 seats, where BUY.md orders six of each. Most of the difference is explained — a

link half carries a pocket at both ends and is printed four times, so a part reused at four positions offers more seats than the chain has joints, and six joints at two bearings each is the 12 that are actually loaded. What is not yet established is which seats carry the load at a shared joint, where the outboard end of one link and the inboard end of the next both present a pocket. Until that is settled the assembly shows a bearing in every seat: geometrically honest, and three more than will be bought.

The Ø37 count is the one to resolve first, because it runs the other way — four seats against six ordered.

24.3 Two things the render made visible

Seating a bearing in every pocket put two bearings inside each other at the elbow. At a shared joint the outboard pocket of one link and the inboard pocket of the next are the same hole in space, so J3 received four bearings occupying two locations. Seats are now deduplicated by position — one bearing per hole — which takes the assembly from 15 bearings to 13.

J4 carries bearings but no shaft. A joint shaft is placed at each link's inboard end, so L2 puts one at J2 and L3 puts one at J3; nothing is placed at the outboard end of the forearm. Neither assembly has ever placed one there, but until the bearings were drawn the gap sat inside a hollow link end and was invisible. Rendered, J4 is a bright ring around an empty hole.

Either the forearm's outboard Ø42 pockets are spare seats — the link half is one part printed four times and carries a pocket at both ends, and the wrist mounts on its own Ø37 bearings — or J4 is a real joint missing its shaft. That is the same open question as the seat census, seen from the other side, and it is a design decision rather than a modelling one.

A smaller point, and not a defect: at J2 and J3, where a shaft is fitted, a small hole remains visible down the joint axis. That is the Ø26 bore of the Ø30 shaft. The M5 x 90 through-bolt that fills it is not modelled — fasteners in this project are accounted for by mass, not geometry.